Chapter 5. The Strong Markov Property

Stopping times and filtrations, Markov processes, the reflection principle, path properties of Brownian motion

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5.1 Stopping times and filtrations

Motivation. Restarting at a random time chosen without looking ahead: stopping times, $\mathcal{F}_T$, and measurability of X_T.

Let $(\Omega, \mathcal{F}, P)$ be a probability space with a **filtration** $(\mathcal{F}_t)_{t\geq 0}$ ($\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$ for $s \leq t$); $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$ is a **filtered probability space**. $(\mathcal{F}_t)$ is **right continuous** if $\mathcal{F}_t = \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}$ for all t. It satisfies the **usual conditions** if it is right continuous, $(\Omega, \mathcal{F}, P)$ is complete, and every P-null set of $\mathcal{F}$ lies in $\mathcal{F}_0$; then all $(\Omega, \mathcal{F}_t, P)$ are complete.

Remark 5.1

Every probability space can be completed. With $\mathcal{N}_P := \{A \subset \Omega \mid A \subset B \in \mathcal{F},\ P(B) = 0\}$, the **completion** $\mathcal{F}^P := \{C \cup A \mid C \in \mathcal{F},\ A \in \mathcal{N}_P\} = \sigma(\mathcal{F}, \mathcal{N}_P)$ is a σ-algebra, and $P(C \cup A) := P(C)$ extends P to it. [Supplement] *Reason:* $\mathcal{N}_P$ is closed under countable unions; $(C \cup A)^c = (C^c \setminus B) \cup (C^c \cap B \setminus A)$ for null $B \supset A$; representations differ by null subsets, and σ-additivity is inherited.

A process $X = (X_t)$ with values in a metric space S is **adapted** to $(\mathcal{F}_t)$ if each X_t is $\mathcal{F}_t$-measurable. Its **natural filtration** $\mathcal{F}_t^0 = \mathcal{F}_t^0(X) := \sigma(X_s \mid s \leq t)$ (also $\mathcal{F}_t^X$) is the smallest filtration making X adapted. X is **continuous** if $P(t \mapsto X_t \text{ continuous}) = 1$, and it has **continuous sample paths** if every path is continuous.

Example 5.2

In the canonical d-dimensional BM $\mathbb{M}_W = (C(\mathbb{R}_+, \mathbb{R}^d), \mathcal{B}(C(\mathbb{R}_+, \mathbb{R}^d)), (W_t), P_x)$, the filtration $(\mathcal{F}_t^{0,+})$ is right continuous and W has continuous sample paths. For a general d-dimensional BM $(\Omega, \mathcal{F}, (W_t), P_x)$ as in Definition 4.18, let $\mathcal{F}_t^0 := \sigma(W_s \mid s \leq t)$, $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0$, $\mathcal{N}_x := \mathcal{N}_{P_x}$, and

$$\mathcal{F}_t^{0,+,x} := \sigma(\mathcal{F}_t^{0,+}, \mathcal{N}_x), \qquad \mathcal{F}_t^{0,x} := \sigma(\mathcal{F}_t^0, \mathcal{N}_x),$$
$$\mathcal{F}_t^{0,x,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,x}, \qquad \mathcal{F}_\infty^{0,+,x} := \sigma(\mathcal{F}_t^{0,+,x} \mid t \geq 0).$$

Then $\mathcal{F}_t^{0,+,x} = \mathcal{F}_t^{0,x,+}$: completing and making right continuous commute (Rogers–Williams, Vol. 1, II.(67.4)). So $(\Omega, \mathcal{F}_\infty^{0,+,x}, (\mathcal{F}_t^{0,+,x}), P_x)$ satisfies the usual conditions and W is continuous. For the canonical model, as Remark 4.15 suggests (Blumenthal's law holds in d dimensions), even $\mathcal{F}_t^{0,x}$ is right continuous (Karatzas–Shreve, Prop. 2.7.7), so $\mathcal{F}_t^{0,x} = \mathcal{F}_t^{0,x,+} = \mathcal{F}_t^{0,+,x}$. Also $\mathcal{F}_t := \bigcap_{x\in\mathbb{R}^d} \mathcal{F}_t^{0,+,x}$, $\mathcal{F}_\infty := \sigma(\mathcal{F}_t \mid t \geq 0)$: right continuous, differing from $(\mathcal{F}_t^{0,+})$ by sets null for every P_x, so the Markov property holds for it too. Continuous paths plus a right continuous filtration settle the measurability questions below.

Definition 5.3 (Stopping time)

$T : \Omega \to [0, \infty]$ is an $(\mathcal{F}_t)$**-stopping time** if

$$\{T \leq t\} \in \mathcal{F}_t \qquad \forall t \geq 0. \tag{5.1}$$

In particular T is $\mathcal{F}$-measurable.

Lemma 5.4

If $(\mathcal{F}_t)$ is right continuous, (5.1) is equivalent to

$$\{T < t\} \in \mathcal{F}_t \qquad \forall t > 0. \tag{5.2}$$

(A T with (5.2) is an **optional** or **weak stopping time**.)

Proof. For a stopping time, $\{T < t\} = \bigcup_n \{T \le t - \frac{1}{n}\} \in \mathcal{F}_t$. Conversely, under (5.2), $\{T \le t\} = \bigcap_{k \ge n} \{T < t + \frac{1}{k}\} \in \mathcal{F}_{t+\frac{1}{n}}$ for every n, hence $\{T \le t\} \in \bigcap_n \mathcal{F}_{t+\frac{1}{n}} = \mathcal{F}_t$ by right continuity. ∎

Example 5.5

(a) If S, T are stopping times, so are $S \vee T$, $S \wedge T$, $S + T$: $\{S \vee T \le t\} = \{S \le t\} \cap \{T \le t\}$ and $\{S \wedge T \le t\} = \{S \le t\} \cup \{T \le t\}$. For the sum, $\tilde{T} \wedge t$ is $\mathcal{F}_t$-measurable for every stopping time $\tilde{T}$ (Lemma 5.6(ii)), and

$$\{S + T \le t\} = \{S \wedge t + \mathbb{1}_{\{S > t\}} + T \wedge t + \mathbb{1}_{\{T > t\}} \le t\} \in \mathcal{F}_t.$$

(b) For stopping times T_n, $\sup_n T_n$ is a stopping time ($\{\sup_n T_n \le t\} = \bigcap_n \{T_n \le t\}$). $\inf_n T_n$, $\liminf_n T_n$, $\limsup_n T_n$ are in general only weak stopping times ($\{\inf_n T_n < t\} = \bigcup_n \{T_n < t\}$, $\limsup_n T_n = \inf_k \sup_{n \ge k} T_n$, $\liminf_n T_n = \sup_k \inf_{n \ge k} T_n$); they are stopping times if $(\mathcal{F}_t)$ is right continuous.

(c) (*Hitting times*) Let X be adapted with values in a metric space (S, d), and assume either (i) X has continuous sample paths and $(\mathcal{F}_t)$ is right continuous, or (ii) X is continuous and $(\mathcal{F}_t)$ satisfies the usual conditions. Under (ii) we may assume (i), since the null set of discontinuous paths and all its subsets lie in $\mathcal{F}_0$. For $A \subset S$ let

$$\sigma_A := \inf\{t > 0 \mid X_t \in A\} \text{ (first hitting time)},$$
$$D_A := \inf\{t \ge 0 \mid X_t \in A\} \text{ (first entry time)},$$

with $\inf \emptyset := \infty$, so $\{\sigma_A \ge t\} = \{X_s \notin A \ \forall s \in (0, t)\}$ and $\{D_A \ge t\} = \{X_s \notin A \ \forall s \in [0, t)\}$.

A open: by continuity $\sigma_A = D_A$, and since A^c is closed, $\{\sigma_A \ge t\} = \bigcap_{s \in [0,t) \cap \mathbb{Q}} \{X_s \notin A\} \in \mathcal{F}_t$.

A closed: $s \mapsto d(X_s, A)$ is continuous, so it is bounded below by a positive constant on $[0, t]$ whenever it is positive there, and

$$\{D_A > t\} = \{d(X_s, A) > 0 \ \forall s \in [0, t]\} = \bigcup_n \bigcap_{s \in [0,t] \cap \mathbb{Q}} \{d(X_s, A) \ge \tfrac{1}{n}\} \in \mathcal{F}_t.$$

[Supplement] Similarly $\{\sigma_A \ge t\} = \bigcap_m \bigcup_n \bigcap_{s \in \mathbb{Q} \cap [\frac{1}{m}, t - \frac{1}{m}]} \{d(X_s, A) \ge \frac{1}{n}\} \in \mathcal{F}_t$.

In all cases σ_A, D_A are weak stopping times, hence stopping times by Lemma 5.4. (For general Borel A and right continuous paths this remains true under extra conditions on S; the proof is hard, see Blumenthal–Getoor, Ch. I, (10.7).)

(d) (*Dyadic approximation*) For a stopping time T let

$$T_n := \sum_{k \ge 0} (k+1) 2^{-n} \mathbb{1}_{[k2^{-n}, (k+1)2^{-n})}(T) \ \text{ on } \{T < \infty\}, \qquad T_n := \infty \ \text{ on } \{T = \infty\}.$$

Then $T_n \searrow T$ and the T_n are stopping times: for $d \in D_n := \{k2^{-n} \mid k \ge 1\}$, $\{T_n = d\} = \{T < d\} \cap \{T \ge d - 2^{-n}\} \in \mathcal{F}_d$, and $\{T_n \le t\} = \bigcup_{d \in D_n,\, d \le t} \{T_n = d\} \in \mathcal{F}_t$.

For a stopping time T, the **σ-algebra of the T-past** is

$$\mathcal{F}_T := \{A \in \mathcal{F} \mid A \cap \{T \le t\} \in \mathcal{F}_t \ \forall t \ge 0\}.$$

[Supplement] It is a σ-algebra: $\Omega \cap \{T \le t\} \in \mathcal{F}_t$; $A^c \cap \{T \le t\} = \{T \le t\} \setminus (A \cap \{T \le t\})$; unions are clear. For $T \equiv t$, $\mathcal{F}_T = \mathcal{F}_t$. T is $\mathcal{F}_T$-measurable, since $\{T \le s\} \cap \{T \le t\} = \{T \le s \wedge t\} \in \mathcal{F}_t$. If $(\mathcal{F}_t)$ is right continuous, then $\mathcal{F}_T = \{A \in \mathcal{F} \mid A \cap \{T < t\} \in \mathcal{F}_t \ \forall t\}$, by the argument of Lemma 5.4 applied to $A \cap \{T \le t\}$.

Lemma 5.6

(i) For stopping times $S \le T$: $\mathcal{F}_S \subset \mathcal{F}_T$.

(ii) For a stopping time T and $t \ge 0$, $T \wedge t$ is $\mathcal{F}_t$-measurable.

(iii) If T_n are decreasing stopping times w.r.t. a right continuous filtration and $T = \inf_n T_n$, then $\mathcal{F}_T = \bigcap_n \mathcal{F}_{T_n}$.

Proof. (i) For $A \in \mathcal{F}_S$: $\{T \le t\} \subset \{S \le t\}$, so $A \cap \{T \le t\} = (A \cap \{S \le t\}) \cap \{T \le t\} \in \mathcal{F}_t$.

(ii) $\{T \wedge t \le t'\}$ is Ω if $t \le t'$, and $\{T \le t'\} \in \mathcal{F}_{t'} \subset \mathcal{F}_t$ if $t' < t$.

(iii) $\mathcal{F}_T \subset \bigcap_n \mathcal{F}_{T_n}$ by (i). Conversely, if $A \in \bigcap_n \mathcal{F}_{T_n}$, then $A \cap \{T_n < t\} \in \mathcal{F}_t$ for all n (right continuity), so $A \cap \{T < t\} = A \cap \bigcup_n \{T_n < t\} \in \mathcal{F}_t$ for all t, i.e. $A \in \mathcal{F}_T$. ■

For $T(\omega) < \infty$ put $X_T(\omega) := X_{T(\omega)}(\omega)$; for a finite stopping time this defines X_T on Ω. For instance, $T \wedge t$ is a finite stopping time.

Lemma 5.7

Let X be adapted with values in a metric space, under the conditions of Example 5.5(c)(i) or (ii), and T a finite stopping time. Then X_T is $\mathcal{F}_T$-measurable. In particular $X_{t\wedge T}$ is $\mathcal{F}_{t\wedge T} \subset \mathcal{F}_t$-measurable, so $(X_{t\wedge T})_{t\ge 0}$ is adapted.

Proof.

Step 1 (dyadic times). Take $T_n \searrow T$ as in Example 5.5(d) and $B \in \mathcal{B}(S)$. $\{X_{T_n} \in B\} = \bigcup_k \{T_n = (k+1)2^{-n}\} \cap \{X_{(k+1)2^{-n}} \in B\} \in \mathcal{F}$, and

$$\{X_{T_n} \in B\} \cap \{T_n \le t\} = \bigcup_{d \in D_n,\, d \le t} \{X_d \in B\} \cap \{T_n = d\} \in \mathcal{F}_t,$$

so $X_{T_n} \in \mathcal{F}_{T_n}$.

Step 2 (limit). With continuous paths, $X_{T_n} \to X_T$ pointwise. For each n, $X_T = \lim_{k \ge n} X_{T_k}$ is $\mathcal{F}_{T_n}$-measurable (as $\mathcal{F}_{T_k} \subset \mathcal{F}_{T_n}$ for $k \ge n$), so $X_T \in \bigcap_n \mathcal{F}_{T_n} = \mathcal{F}_T$ by Lemma 5.6(iii). ■

Now let X take values in a Borel set $S \subset \mathbb{R}^d$ with $0 \in S$. For a (not necessarily finite) stopping time T define $\mathbb{1}_{\{T<\infty\}} X_T := X_T$ on $\{T < \infty\}$ and $:= 0$ on $\{T = \infty\}$.

Lemma 5.8

Under the assumptions of Lemma 5.7 (with $S \subset \mathbb{R}^d$ Borel, $0 \in S$), $\mathbb{1}_{\{T<\infty\}} X_T$ is $\mathcal{F}_T$-measurable.

Proof. By Lemmas 5.6(i) and 5.7, $\mathbb{1}_{\{T<\infty\}} X_{T\wedge n} \in \mathcal{F}_T$ for each n, and $\mathbb{1}_{\{T<\infty\}} X_{T\wedge n} \to \mathbb{1}_{\{T<\infty\}} X_T$ pointwise. ■

Remark 5.9

Example 5.5(c) remains true with "continuous" replaced by "right continuous". Since $T_n \searrow T$ in Example 5.5(d), Lemmas 5.7 and 5.8 hold for right continuous paths as well.

5.2 Markov processes and the strong Markov property

Motivation. Right continuous paths and the Feller property extend the Markov property to stopping times.

Throughout, $S \subset \mathbb{R}^d$ is Borel with $0 \in S$, equipped with $\mathcal{B}(S)$.

Definition 5.10 (Markovian semigroup)

Maps $P_t : S \times \mathcal{B}(S) \to [0,1]$, $t \geq 0$, form a **markovian semigroup of transition kernels** if (i) $P_t(x,\cdot)$ is a probability measure for all t, x; (ii) $x \mapsto P_t(x,A)$ is $\mathcal{B}(S)$-measurable; (iii) (Chapman–Kolmogorov) $P_t P_s = P_{t+s}$, i.e. $\int_S P_s(y,A)\, P_t(x,dy) = P_{t+s}(x,A)$; (iv) $P_0(x,\cdot) = \delta_x$. For bounded (or nonnegative) measurable f put

$$P_t f(x) := \int_S f(y)\, P_t(x,dy), \tag{5.3}$$

so that $P_t \mathbb{1}_A(x) = P_t(x,A)$.

Definition 5.11 (Normal Markov process)

$\mathbb{M} = (\Omega, \mathcal{F}, (X_t)_{t\geq 0}, (P_x)_{x\in S})$ with state space $(S, \mathcal{B}(S))$ is a **normal Markov process** w.r.t. a filtration $(\mathcal{F}_t)$ if

(M1) $x \mapsto P_x(F)$ is $\mathcal{B}(S)$-measurable for every $F \in \mathcal{F}$;

(M2) (X_t) is $(\mathcal{F}_t)$-adapted and $P_x(X_{s+t} \in A \mid \mathcal{F}_s) = P_{X_s}(X_t \in A)$ P_x-a.s. for all $s,t \geq 0$, $A \in \mathcal{B}(S)$, $x \in S$ (Markov property), where $P_x(\cdot \mid \mathcal{F}_s) = \mathbb{E}_x[\mathbb{1}_A(X_{s+t}) \mid \mathcal{F}_s]$;

(M3) $P_x(X_0 = x) = 1$ for all x (normality).

Theorem 5.12

(i) Let $\mathbb{M} = (\Omega, \mathcal{F}, (X_t), (P_x)_{x\in S})$ with $\mathcal{F} = \sigma(X_t \mid t \geq 0)$.

(a) If there is a markovian semigroup (P_t) with, for all $0 \leq t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(S)$, $t_0 := 0$, $x_0 := x$,

$$P_x((X_{t_1}, \ldots, X_{t_n}) \in A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} P_{t_i - t_{i-1}}(x_{i-1}, dx_i), \tag{5.4}$$

then $\mathbb{M}$ is a normal Markov process w.r.t. $(\mathcal{F}_t^0) = (\sigma(X_s \mid s \leq t))$.

(b) If $\mathbb{M}$ is a normal Markov process, then

$$P_t(x,A) := \mathbb{E}_x[\mathbb{1}_A(X_t)] = P_x(X_t \in A) \tag{5.5}$$

is a markovian semigroup and (5.4) holds.

(ii) For every markovian semigroup there is a normal Markov process satisfying (5.5), with $\Omega = S^{[0,\infty)}$.

Proof. (i)(a) (M3) is (5.4) with $n = 1$, $t_1 = 0$. (M1) holds for cylinder sets by (5.4) and Definition 5.10(ii) (iterated kernel integrals of measurable functions are measurable), and extends to $\mathcal{F}$ by a Dynkin argument. (M2) is equivalent to

$$\mathbb{E}_x[\mathbb{1}_A(X_{t+s})\mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{X_s}[\mathbb{1}_A(X_t)]\mathbb{1}_C] \qquad \forall C \in \mathcal{F}_s^0.$$

For $s = 0$ this is clear ($\mathcal{F}_0^0 = \sigma(X_0)$ is P_x-trivial). For $s > 0$, by the monotone class theorem it suffices to take $C = \{(X_{s_1}, \dots, X_{s_m}) \in C_1 \times \cdots \times C_m\}$, $0 \le s_1 < \cdots < s_m = s$. With $s_{m+1} := t + s$, $y_0 := x$, (5.4) gives

$$\begin{aligned}\mathbb{E}_x[\mathbb{1}_A(X_{t+s})\mathbb{1}_C] &= \int_{C_1} \cdots \int_{C_m} \int_A \prod_{j=1}^{m+1} P_{s_j - s_{j-1}}(y_{j-1}, dy_j) \\ &= \int_{C_1} \cdots \int_{C_m} \mathbb{E}_{y_m}[\mathbb{1}_A(X_t)] \prod_{j=1}^{m} P_{s_j - s_{j-1}}(y_{j-1}, dy_j),\end{aligned}$$

which by (5.4) again equals $\mathbb{E}_x[\mathbb{E}_{X_s}[\mathbb{1}_A(X_t)]\mathbb{1}_C]$.

(i)(b) Definition 5.10(i), (ii),

(iv) are clear from (M1), (M3). By (5.5) and (5.3), $P_t\mathbb{1}_A(x) = \mathbb{E}_x[\mathbb{1}_A(X_t)]$, and by linearity and monotone convergence

$$P_t f(x) = \mathbb{E}_x[f(X_t)] \qquad \text{for bounded or nonnegative measurable } f. \tag{5.6}$$

Chapman–Kolmogorov: with (M2) and (5.6),

$$P_{t+s}(x, A) = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A(X_{t+s}) \mid \mathcal{F}_s]] = \mathbb{E}_x[P_t(X_s, A)] = P_s(P_t\mathbb{1}_A)(x) = \int P_t(y, A)\, P_s(x, dy).$$

For (5.4), it suffices to prove

$$\mathbb{E}_x[f_1(X_{t_1}) \cdots f_n(X_{t_n})] = \int \cdots \int f_1(x_1) \cdots f_n(x_n) \prod_{i=1}^{n} P_{t_i - t_{i-1}}(x_{i-1}, dx_i) \tag{5.7}$$

for bounded measurable f_i, by induction on n; $n = 1$ is (5.6). Conditioning on $\mathcal{F}_{t_{n-1}}$, Theorem 2.35(iii) and (M2) give

$$\mathbb{E}_x\Big[\prod_{i \le n} f_i(X_{t_i})\Big] = \mathbb{E}_x\Big[\prod_{i \le n-1} f_i(X_{t_i})\, P_{t_n - t_{n-1}} f_n(X_{t_{n-1}})\Big],$$

and the induction hypothesis applied to $f_1, \dots, f_{n-2}$, $f_{n-1} \cdot P_{t_n - t_{n-1}} f_n$ yields (5.7).

(ii) The family (5.4) is consistent by Chapman–Kolmogorov. Kolmogorov's extension theorem, whose proof (Theorem 4.7) works verbatim for $\mathbb{R}^d$-valued coordinates (see also Le Gall, Thm. 6.3; Revuz–Yor, Ch. I, (3.2)), gives P_x on the path space. (M1) follows as in (i)(a), and (i)(a) does the rest. ∎

The canonical model. Let $\Omega \subset S^{[0,\infty)}$ (e.g. $S^{[0,\infty)}$ or $C([0,\infty), S)$), $X_t(\omega) := \omega(t)$, $\mathcal{F} := \sigma(X_t \mid t \ge 0)$, $\mathcal{F}_t^0 := \sigma(X_s \mid s \le t)$, and let P_x be probability measures with $P_x(X_0 = x) = 1$. Then $\mathbb{M} = (\Omega, \mathcal{F}, (X_t), (P_x))$ is the **canonical model** of (X_t). The shifts $\theta_s(\omega)(t) := \omega(s + t)$ are $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable with $\hat{\mathcal{F}}_t^0 := \sigma(X_s \mid s \ge t)$. If $\mathbb{M}$ is a normal Markov process w.r.t. $(\mathcal{F}_t)$ (hence also w.r.t. $(\mathcal{F}_t^0)$), we call $\mathbb{M} = (\Omega, \mathcal{F}, (\mathcal{F}_t), (X_t), (P_x), (\theta_t))$ a **canonical Markov process**.

Lemma 5.13

Let the canonical model satisfy (M1). Then (M2) holds w.r.t. $\mathcal{F}_t := \mathcal{F}_t^{0,+} = \bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon}^0$ if and

only if

$$\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s] = \mathbb{E}_{X_s}[Y] \qquad P_x\text{-a.s., } \forall s\ge 0,\ x\in S,\ Y\in\mathcal{F}_b. \tag{5.8}$$

As in Remark 4.12, (5.8) implies $\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s] = \mathbb{E}_x[Y\circ\theta_s \mid \sigma(X_s)]$.

Proof. (M3) holds by definition of the canonical model. "⇐": take $Y=\mathbb{1}_A(X_t)$. "⇒" [Supplement]: by Theorem 5.12(i)(b), (5.4) holds with the semigroup (5.5). For $Y=\prod_{i\le n} f_i(X_{t_i})$, $0\le t_1<\cdots<t_n$, condition successively on $\mathcal{F}_{s+t_{n-1}}, \mathcal{F}_{s+t_{n-2}},\dots$ and use (M2) and Theorem 2.35(iii),(iv), exactly as in the induction for (5.7). The result is

$$\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s] = \Big[P_{t_1}\big(f_1 P_{t_2-t_1}(f_2\cdots P_{t_n-t_{n-1}} f_n)\big)\Big](X_s),$$

and by (5.7) the bracket evaluated at y is $\mathbb{E}_y[Y]$. The class of $Y\in\mathcal{F}_b$ satisfying (5.8) is an MVS containing these products, so Theorem 2.29 gives (5.8). ∎

For a stopping time T put $(\theta_T\omega)(t):=\omega(T(\omega)+t)$ on $\{T<\infty\}$. (In Markov process theory one sets $\theta_T\omega\equiv\Delta$ on $\{T=\infty\}$, with a cemetery point Δ and $f(\Delta):=0$.)

Definition 5.14 (Strong Markov property)

A canonical Markov process $\mathbb{M}$ with right continuous paths and right continuous filtration $(\mathcal{F}_t)$ has the **strong Markov property** (SMP) w.r.t. $(\mathcal{F}_t)$ if for every $(\mathcal{F}_t)$-stopping time T,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} Y\circ\theta_T \mid \mathcal{F}_T] = \mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y] \qquad P_x\text{-a.s., } \forall Y\in\mathcal{F}_b,\ x\in S.$$

The right side, $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{\mathbb{1}_{\{T<\infty\}}X_T}[Y]$, is $\mathcal{F}_T$-measurable by Lemma 5.8, (M1) and Remark 5.9. For finite T the SMP reads $\mathbb{E}_x[Y\circ\theta_T \mid \mathcal{F}_T]=\mathbb{E}_{X_T}[Y]$.

Theorem 5.15

Let $\mathbb{M}=(\Omega,\mathcal{F}=\sigma(X_t\mid t\ge 0),(\mathcal{F}_t),(X_t),(P_x),(\theta_t))$ be a canonical Markov process w.r.t. $\mathcal{F}_t:=\mathcal{F}_t^{0,+}$ such that (i) $\mathbb{M}$ has right continuous sample paths, and (ii) the semigroup (5.5) has the **Feller property** $P_t(C_b(S))\subset C_b(S)$. Then $\mathbb{M}$ has the SMP w.r.t. $(\mathcal{F}_t)$.

Proof.

Step 1 (reduction). Fix x. By the monotone class theorem it suffices to take $Y=f_1(X_{t_1})\cdots f_m(X_{t_m})$ with $f_i\in C_b(S)$, $0\le t_1<\cdots<t_m$. Let $T_n\searrow T$ be the dyadic approximations of Example 5.5(d) and $C\in\mathcal{F}_T$.

Step 2 (discretise). By right continuity of paths, $Y\circ\theta_{T_n}=\prod_i f_i(X_{t_i+T_n})\to Y\circ\theta_T$ on $\{T<\infty\}$, so by the DCT

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} Y\circ\theta_T \mathbb{1}_C] = \lim_n \sum_{k\ge 1} \mathbb{E}_x\big[\mathbb{1}_{C\cap\{(k-1)2^{-n}\le T<k2^{-n}\}}\, Y\circ\theta_{k2^{-n}}\big].$$

Here $C\cap\{(k-1)2^{-n}\le T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$, since $C\cap\{T<k2^{-n}\}\in\mathcal{F}_{k2^{-n}}$ by right continuity and $\{T\ge(k-1)2^{-n}\}\in\mathcal{F}_{(k-1)2^{-n}}$.

Step 3 (Markov at fixed times). By (5.8) (Lemma 5.13) each summand equals

$$\mathbb{E}_x\big[\mathbb{1}_{C\cap\{(k-1)2^{-n}\le T<k2^{-n}\}}\,\mathbb{E}_{X_{k2^{-n}}}[Y]\big],$$

so the sum is $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_{T_n}}[Y]\mathbb{1}_C]$.

Step 4 (Feller). $x\mapsto\mathbb{E}_x[Y]=P_{t_1}(f_1P_{t_2-t_1}(f_2\cdots P_{t_m-t_{m-1}}f_m))(x)$ is continuous by the Feller property, and $X_{T_n}\to X_T$ on $\{T<\infty\}$. The DCT gives the limit $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y]\mathbb{1}_C]$. ∎

Time-dependent functionals. Let $Y : \mathbb{R} \times \Omega \to \mathbb{R}$ be bounded and $\mathcal{B}(\mathbb{R}) \otimes \mathcal{F}$-measurable, and define $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]$ as

$$\omega \mapsto \mathbb{1}_{\{T<\infty\}}(\omega)\, \mathbb{E}_{\mathbb{1}_{\{T<\infty\}}X_T(\omega)}[Y(T(\omega),\cdot)], \tag{5.9}$$

and $(Y(T,\cdot)\circ\theta_T)(\omega) := Y(T(\omega),\theta_T\omega)$ (the shift acts only on the second variable). If Y does not depend on time, these reduce to $\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y]$ and $Y\circ\theta_T$.

Definition 5.16 (Extended strong Markov property)

$\mathbb{M}$ as in Definition 5.14 has the **extended strong Markov property** (ESMP) w.r.t. $(\mathcal{F}_t)$ if for every stopping time T, $Y \in (\mathcal{B}(\mathbb{R})\otimes\mathcal{F})_b$ and $x \in S$,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T,\cdot)\circ\theta_T \mid \mathcal{F}_T] = \mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)] \qquad P_x\text{-a.s.}$$

[Supplement] The right side (5.9) is $\mathcal{F}_T$-measurable. For Y as in (5.10) below it equals $\mathbb{1}_{\{T<\infty\}}\Phi(T,X_T)$ with Φ continuous (Feller), and T, $\mathbb{1}_{\{T<\infty\}}X_T$ are $\mathcal{F}_T$-measurable (Lemma 5.8). The monotone class argument at the end of the next proof extends this to all bounded Y.

Theorem 5.17

Under the assumptions of Theorem 5.15 ($\mathcal{F}_t = \mathcal{F}_t^{0,+}$, right continuous paths, Feller property), $\mathbb{M}$ has the ESMP w.r.t. $(\mathcal{F}_t)$.

Proof.

Step 1 (product functionals). Fix x and first let

$$Y(s,\omega) = g(s)f_1(X_{t_1}(\omega))\cdots f_m(X_{t_m}(\omega)), \qquad g \in C_b(\mathbb{R}),\ f_i \in C_b(S). \tag{5.10}$$

With T_n, $C \in \mathcal{F}_T$ as before, right continuity and the DCT give $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T,\theta_T)\mathbb{1}_C] = \lim_n \mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}Y(T_n,\theta_{T_n})\mathbb{1}_C]$. Splitting according to $\{(k-1)2^{-n} \le T < k2^{-n}\} \in \mathcal{F}_{k2^{-n}}$, on which $Y(T_n,\cdot) = Y(k2^{-n},\cdot) \in \mathcal{F}_b$, the Markov property at the fixed times $k2^{-n}$ turns this into $\lim_n \mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_{T_n}}[Y(T_n,\cdot)]\mathbb{1}_C]$. Since $(s,y) \mapsto \mathbb{E}_y[Y(s,\cdot)] = g(s)P_{t_1}(f_1P_{t_2-t_1}(\cdots P_{t_m-t_{m-1}}f_m))(y)$ is continuous (Feller), the DCT gives $\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}}\mathbb{E}_{X_T}[Y(T,\cdot)]\mathbb{1}_C]$.

Step 2 (monotone class). The $Y \in (\mathcal{B}(\mathbb{R})\otimes\mathcal{F})_b$ satisfying the ESMP form an MVS. The multiplicative class $\mathcal{M}$ of indicators of $U_0 \times \{(X_{t_1},\dots,X_{t_m}) \in U_1\times\cdots\times U_m\}$, U_j open, generates $\mathcal{B}(\mathbb{R})\otimes\mathcal{F}$. Each such indicator is an increasing limit of functions of the form (5.10) (use $\phi(m\,d(\cdot,U^c))$-type cut-offs as in Theorem 3.35, (ii)⇒(iii)), so $\mathcal{M} \subset \mathcal{H}$ and Theorem 2.29 concludes. ∎

Example 5.18

The canonical d-dimensional Brownian motion $\mathbb{M}_W$ has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$ (Section 4.4), continuous paths, and the Feller property: $P_tf(x) = \int f(y)p_t(x,y)\,dy$ is continuous in x for $f \in C_b$ by the DCT. So it has the SMP and ESMP w.r.t. $(\mathcal{F}_t^{0,+})$ and $(\mathcal{F}_t)$ (Example 5.2).

Remark 5.19

Going through the proofs of Theorems 5.15, 5.17 (cf. Remarks 4.17, 4.19), a d-dimensional BM as in Definition 4.18 satisfies the ESMP for fixed x w.r.t. $(\mathcal{F}_t^{0,+,x})$ (needed for measurability, as it satisfies the usual conditions), for Y of the form (5.10) with $Y(T,\cdot)\circ\theta_T$ replaced by

$g(T)f_1(W_{t_1+T})\cdots f_m(W_{t_m+T})$. In particular, for $f \in \mathcal{B}(\mathbb{R}^d)_b$ and every $(\mathcal{F}_t^{0,+,x})$-stopping time T,

$$\mathbb{E}_x[\mathbb{1}_{\{T<\infty\}} f(W_{t+T}) \mid \mathcal{F}_T^{0,+,x}] = \mathbb{1}_{\{T<\infty\}} \mathbb{E}_{W_T}[f(W_t)] \qquad P_x\text{-a.s.} \tag{5.11}$$

If T is an $(\mathcal{F}_t^{0,+})$-stopping time (hence an $(\mathcal{F}_t^{0,+,x})$-stopping time for every x), (5.11) holds simultaneously for all x.

5.3 Applications to Brownian motion

Motivation. Restart at hitting times plus symmetry: reflection principle, maximum, arcsine law; time inversion for 0 and ∞.

5.3.1 The reflection principle for one-dimensional BM

Let $\mathbb{M}_B = (\Omega, \mathcal{F}, (\mathcal{F}_t^{0,+}), (B_t), P_0, (\theta_t))$ be the canonical one-dimensional BM; it has the ESMP (Example 5.18). For $a, t > 0$ we show

$$P_0(D_{\{a\}} < t,\ B_t > a) = \tfrac{1}{2} P_0(D_{\{a\}} < t). \tag{5.12}$$

Let $Y(s,\omega) := \mathbb{1}_{(-\infty,t)}(s)\mathbb{1}_{(a,\infty)}(B_{(t-s)^+}(\omega))$ (bounded, $\mathcal{B}(\mathbb{R}) \otimes \mathcal{F}$-measurable) and $T := \inf\{0 \le s < t \mid B_s = a\}$ (a stopping time, Example 5.5(c)). Then $\{T < \infty\} = \{T < t\} = \{D_{\{a\}} < t\}$ and $\mathbb{1}_{\{T<\infty\}} Y(T, \theta_T) = \mathbb{1}_{\{D_{\{a\}}<t\}} \mathbb{1}_{\{B_t > a\}}$. Taking expectations in the ESMP (Theorem 5.17), and using $B_T = a$ and $\mathbb{E}_a[\mathbb{1}_{\{B_r>a\}}] = \frac{1}{2}$ for $r > 0$ (symmetry),

$$P_0(D_{\{a\}} < t,\ B_t > a) = \mathbb{E}_0[\mathbb{1}_{\{T<t\}} \mathbb{E}_{B_T}[\mathbb{1}_{\{B_{t-T}>a\}}]] = \tfrac{1}{2} P_0(T < t),$$

which is (5.12). Since $B_0 = 0 < a$, path continuity gives $\{B_t > a\} \subset \{D_{\{a\}} < t\}$, hence

$$P_0(D_{\{a\}} < t) = 2P_0(B_t > a), \tag{5.13}$$

and, by continuity from above,

$$P_0(D_{\{a\}} \le t) = \lim_n P_0(D_{\{a\}} < t + \tfrac{1}{n}) = 2 \lim_n P_0(B_{t+\frac{1}{n}} > a) = \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-\frac{x^2}{2t}}\, dx = 2P_0(B_t \ge a). \tag{5.14}$$

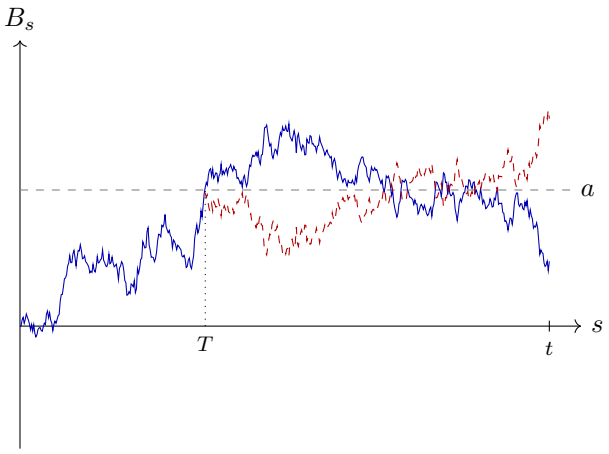


Figure. *Reflection after $T = D_{\{a\}}$:* $P_0(D_{\{a\}} < t) = 2P_0(B_t > a)$.

Proposition 5.20

(i) For $a \in \mathbb{R}\setminus\{0\}$ and $t \geq 0$: $P_0(D_{\{a\}} \leq t) = \int_0^t \frac{|a|}{\sqrt{2\pi s^3}} e^{-\frac{a^2}{2s}}\, ds$. (ii) For $a, t > 0$: $P_0(\sup_{s\leq t} B_s \leq a) = P_0(|B_t| \leq a) = \frac{2}{\sqrt{2\pi t}} \int_0^a e^{-\frac{x^2}{2t}}\, dx$.

Proof. (i) For $a, t > 0$ substitute $x = t^{1/2} a\, s^{-1/2}$ in (5.14):

$$P_0(D_{\{a\}} \leq t) = \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-\frac{x^2}{2t}}\, dx = 2\int_0^t \frac{1}{\sqrt{2\pi t}} \frac{t^{1/2} a}{2 s^{3/2}} e^{-\frac{a^2}{2s}}\, ds = \int_0^t \frac{a}{\sqrt{2\pi s^3}} e^{-\frac{a^2}{2s}}\, ds.$$

For $a < 0$ use that $(-B_t)$ is again a canonical BM; $t = 0$ is trivial.

(ii) For $a, t > 0$, $\{D_{\{a\}} > t\} = \{\sup_{s\leq t} B_s < a\}$. By continuity from above and (5.13),

$$P_0\Big(\sup_{s\leq t} B_s \leq a\Big) = \lim_n P_0(D_{\{a+\frac{1}{n}\}} > t) = 1 - 2\lim_n P_0(B_t \geq a + \tfrac{1}{n}) = 1 - 2P_0(B_t > a) = P_0(|B_t| \leq a).$$

■

Remark 5.21

Proposition 5.20 and the arcsine law below hold for every BM $(\tilde{B}_t)$ on $(\tilde{\Omega}, \tilde{\mathcal{F}}, \tilde{P}_0)$ as in Definition 4.1. A shift-free strong Markov property gives this directly (Le Gall, Ch. 2.4); alternatively, the finite-dimensional distributions agree with those of $\mathbb{M}_B$, and the events involved are determined by countably many times:

$$\tilde{P}_0\Big(\sup_{s\leq t} \tilde{B}_s \leq a\Big) = \tilde{P}_0\Big(\sup_{s\in\mathbb{Q}\cap[0,t]} \tilde{B}_s \leq a\Big) = P_0\Big(\sup_{s\in\mathbb{Q}\cap[0,t]} B_s \leq a\Big) = P_0\Big(\sup_{s\leq t} B_s \leq a\Big).$$

Similarly (with a filtration satisfying the usual conditions) $\tilde{P}_0(\tilde{D}_{\{a\}} \leq t) = P_0(D_{\{a\}} \leq t)$, because $\{D_{\{a\}} > t\} = \bigcup_n \bigcap_{s\in[0,t]\cap\mathbb{Q}} \{|B_s - a| \geq \frac{1}{n}\}$, and likewise for $\tilde{B}$.

Example 5.22 (Arcsine law)

Let $L := \sup\{t \leq 1 \mid B_t = 0\}$, the last zero before time 1. L is not a stopping time, but its law can be computed. For $s \in [0,1)$, $\{L \leq s\} = \{B_t \neq 0\ \forall t \in (s,1]\}$, and $P_0(L \leq 1) = 1$. By the Markov property,

$$P_0(L \leq s) = \mathbb{E}_0[\mathbb{E}_0[\mathbb{1}_{\{B_{t-s}\neq 0\ \forall t\in(s,1]\}} \circ \theta_s \mid \mathcal{F}_s^{0,+}]] = \mathbb{E}_0[f(B_s)] = \int p_s(0,y)\, P_y(\sigma_{\{0\}} > 1-s)\, dy,$$

with $f(y) := P_y(B_t \neq 0\ \forall t \in (0, 1-s])$. Since $P_y(\sigma_{\{0\}} > 1-s) = P_y(D_{\{0\}} > 1-s) = P_0(D_{\{-y\}} > 1-s)$, Proposition 5.20(i) and symmetry give

$$\begin{aligned} P_0(L \leq s) &= 2\int_0^\infty \frac{e^{-y^2/(2s)}}{\sqrt{2\pi s}} \Big(\int_{1-s}^\infty \frac{y}{\sqrt{2\pi r^3}} e^{-\frac{y^2}{2r}}\, dr\Big) dy = \frac{1}{\pi}\int_{1-s}^\infty \frac{1}{\sqrt{sr^3}} \int_0^\infty y e^{-\frac{(r+s)y^2}{2rs}}\, dy\, dr \\ &= \frac{1}{\pi}\int_{1-s}^\infty \frac{1}{\sqrt{sr^3}} \frac{rs}{r+s}\, dr = \frac{1}{\pi}\int_{1-s}^\infty \sqrt{\frac{r}{s}}\, \frac{s}{(r+s)^2}\, dr. \end{aligned}$$

The substitution $t = \frac{s}{r+s}$, $dt = -\frac{s}{(r+s)^2} dr$, gives

$$P_0(L \leq s) = \frac{1}{\pi}\int_0^s \frac{dt}{\sqrt{t(1-t)}} = \frac{2}{\pi} \arcsin\sqrt{s}, \qquad s \in [0,1].$$

The density $\frac{1}{\pi\sqrt{t(1-t)}}$ is largest near 0 and 1: the last zero before time 1 is most likely to occur either very early or very late.

5.3.2 Further path properties of BM

Theorem 5.23 (Time inversion)

If (B_t) is a standard BM, so is $X_t := tB_{1/t}$ for $t > 0$, $X_0 := 0$.

Proof.

Step 1 (Definition 4.1(i), (ii)). (i) is clear. For $0 < s < t$, $0 < \frac{1}{t} < \frac{1}{s}$ and

$$X_t - X_s = (t-s)\underbrace{B_{1/t}}_{=:Y_1} - s\underbrace{(B_{1/s} - B_{1/t})}_{=:Y_2},$$

with independent $Y_1 \sim N(0, \frac{1}{t})$, $Y_2 \sim N(0, \frac{1}{s} - \frac{1}{t})$. By Example 3.44(i), $X_t - X_s \sim N\big(0, (t-s)^2\frac{1}{t} + s^2(\frac{1}{s} - \frac{1}{t})\big) = N(0, t-s)$. The case $s = 0$ is similar.

Step 2 (Definition 4.1(iii)). For $s, t > 0$, $\mathbb{E}[X_s X_t] = st(\frac{1}{s} \wedge \frac{1}{t}) = s \wedge t$, and this also holds if s or t is 0. [Supplement] Increments $Z = (X_{t_1}, X_{t_2} - X_{t_1}, \dots)$ are MG, G independent standard normals (normalised increments of B at $1/t_i$), so by Theorem 3.52 $\varphi_Z(u) = \mathbb{E}[e^{i\langle M^\top u, G\rangle}] = e^{-\frac{1}{2}\|M^\top u\|^2} = e^{-\frac{1}{2}u^\top \Sigma u}$ with $\Sigma = MM^\top$ the covariance matrix of Z. From $\mathbb{E}[X_s X_t] = s \wedge t$, Σ is diagonal with entries $t_i - t_{i-1}$. So $\varphi_Z(u) = \prod_i e^{-\frac{1}{2}(t_i - t_{i-1})u_i^2} = \prod_i \varphi_{Z_i}(u_i)$, and Remark 3.50 gives independence.

Step 3 (continuity). By Steps 1–2, X satisfies (4.9), so X and B have the same finite-dimensional distributions. Paths of X are continuous on $(0, \infty)$ a.s. (everywhere if B is canonical). At 0: the laws of X and B on $\mathbb{Q} \cap (0, \infty)$ agree, and $\{\lim_{t \downarrow 0,\, t \in \mathbb{Q}} X_t = 0\}$ is determined by countably many coordinates, so $P(\lim_{t \downarrow 0,\, t \in \mathbb{Q}} X_t = 0) = P(\lim_{t \downarrow 0,\, t \in \mathbb{Q}} B_t = 0) = 1$. With continuity on $(0, \infty)$ this gives $\lim_{t \downarrow 0} X_t = 0$ a.s. ∎

Remark 5.24

Time inversion links the behaviour of B_t as $t \to \infty$ with that of X_t as $t \to 0$. For example $B_t/t = X_{1/t} \to 0$, so paths grow slower than linearly; Theorem 5.26 says more. From now on $\mathbb{M}_B = (\Omega, \mathcal{F}, (B_t), (P_x), (\theta_t))$ is the canonical one-dimensional BM, $\hat{\mathcal{F}}_t^0 := \sigma(B_s \mid s \geq t)$ (decreasing in t) and

$$\mathcal{T} := \bigcap_{t \geq 0} \hat{\mathcal{F}}_t^0$$

is the **tail σ-algebra** of B: the events of the remote future. For example, $\limsup_n A_n \in \mathcal{T}$ for $A_m \in \hat{\mathcal{F}}_m^0$, since $\bigcup_{m \geq n} A_m \in \hat{\mathcal{F}}_n^0$ and $(\hat{\mathcal{F}}_n^0)$ is decreasing.

Theorem 5.25

P_0 is deterministic on $\mathcal{T}$: $P_0(A) \in \{0, 1\}$ for all $A \in \mathcal{T}$.

Proof.

Step 1 (tail of B = germ of X). With X from Theorem 5.23, using $X_0 \equiv 0$ (so adding it changes no σ-algebra),

$$\begin{aligned}\mathcal{T} &= \bigcap_{t>0} \sigma(B_s \mid s \geq t) = \bigcap_{t>0} \sigma(B_{1/s} \mid 0 < s \leq \tfrac{1}{t}) = \bigcap_{t>0} \sigma(X_s \mid 0 < s \leq t) \\ &= \bigcap_{t>0} \sigma(X_s \mid 0 \leq s \leq t) = \mathcal{F}_0^{0,+}(X).\end{aligned}$$

Step 2 (Blumenthal for X) [Supplement]. Transfer from Theorem 4.16 (lectures: Le Gall, Thm. 2.13). Set $X = 0$ on the null set where it is discontinuous at 0; then $X : \Omega \to C(\mathbb{R}_+)$

is measurable, $P_0 \circ X^{-1} = P_0^W$ (Wiener measure, same finite-dimensional laws), and $\sigma(X_s \mid s \le t) = X^{-1}(\mathcal{F}_t^0)$. For $A \in \mathcal{F}_0^{0,+}(X)$ choose $\Gamma_n \in \mathcal{F}_{1/n}^0$ with $A = X^{-1}(\Gamma_n)$. Then $P_0^W(\Gamma_n \triangle \Gamma_m) = P_0(X^{-1}(\Gamma_n \triangle \Gamma_m)) = 0$, so $\Gamma := \limsup_n \Gamma_n \in \mathcal{F}_0^{0,+}$ differs from each Γ_n by a P_0^W-null set. Hence $P_0(A) = P_0^W(\Gamma) \in \{0,1\}$ by Theorem 4.16. ∎

Theorem 5.26

For the canonical one-dimensional BM, P_0-a.s.

$$\limsup_{t\to\infty} \frac{B_t}{\sqrt{t}} = +\infty, \qquad \liminf_{t\to\infty} \frac{B_t}{\sqrt{t}} = -\infty.$$

Proof. Let $K > 0$. By Fatou applied to indicators and scaling invariance (Proposition 4.2),

$$P_0\Big(\limsup_n \Big\{\frac{B_n}{\sqrt{n}} \ge K\Big\}\Big) \ge \limsup_n P_0\Big(\frac{B_n}{\sqrt{n}} \ge K\Big) = P_0(B_1 \ge K) > 0.$$

The event on the left lies in $\mathcal{T}$, so by Theorem 5.25 it has probability 1. As K is arbitrary, $\limsup_t B_t/\sqrt{t} = \infty$ a.s.; the second claim follows by symmetry ($-B$ is a BM). ∎

The next corollary holds for every BM, but we prove it for the canonical one, as a consequence of the SMP w.r.t. $(\mathcal{F}_t^{0,+})$.

Corollary 5.27

Let $N(\omega) := \{s \in [0,1] \mid B_s(\omega) = 0\}$. For P_0-a.e. ω: (i) $N(\omega)$ is closed;
(ii) $N(\omega)$ has Lebesgue measure zero;
(iii) $N(\omega)$ has no isolated points. So $N(\omega)$ is a Cantor-type set: closed, null, and every point is a cluster point.

Proof. (i) ω is continuous and $\{0\}$ is closed.

(ii) By Fubini, $\mathbb{E}_0\big[\int_0^1 \mathbb{1}_{N(\cdot)}(s)\,ds\big] = \int_0^1 P_0(B_s = 0)\,ds = 0$, so $\lambda(N(\omega)) = 0$ for P_0-a.e. ω.

(iii)

Step 1. Let $A_{r,s} := \{N(\omega) \cap (r,s) \text{ is exactly one point}\}$; then

$$J := \{N(\omega) \text{ has isolated points}\} = \bigcup_{r<s\in\mathbb{Q}_+} A_{r,s}.$$

Let $T_r := r + \sigma_{\{0\}} \circ \theta_r = \inf\{t > r \mid B_t = 0\}$.

Step 2 (T_r is a stopping time) [Supplement]. $\{T_r < t\} = \bigcup_n \{\exists s \in [r + \frac{1}{n}, t - \frac{1}{n}] : B_s = 0\} = \bigcup_n \bigcap_m \bigcup_{q\in\mathbb{Q}\cap[r+\frac{1}{n},t-\frac{1}{n}]} \{|B_q| < \frac{1}{m}\} \in \mathcal{F}_t^0$ (continuity and compactness), so T_r is a weak stopping time, hence an $(\mathcal{F}_t^{0,+})$-stopping time (Lemma 5.4).

Step 3 ($P_0(\sigma_{\{0\}} = 0) = 1$) [Supplement]. $A_+ := \bigcap_n \{\sup_{s\le 1/n} B_s > 0\} \in \mathcal{F}_0^{0,+}$ and $P_0(\sup_{s\le 1/n} B_s > 0) \ge P_0(B_{1/n} > 0) = \frac{1}{2}$, so $P_0(A_+) \ge \frac{1}{2}$ and, by Blumenthal (Theorem 4.16), $P_0(A_+) = 1$. Likewise $A_- := \bigcap_n \{\inf_{s\le 1/n} B_s < 0\}$ has probability 1. On $A_+ \cap A_-$ the path takes both signs in every $(0, \frac{1}{n})$, hence vanishes there (intermediate value theorem), so $\sigma_{\{0\}} = 0$.

Step 4 (SMP). Since $B_{T_r} = 0$ on $\{T_r < \infty\}$, by Theorem 2.35(iii) and the SMP,

$$P_0(A_{r,s}) \le P_0(T_r < s,\ \sigma_{\{0\}} \circ \theta_{T_r} > 0) = \mathbb{E}_0\big[\mathbb{1}_{\{T_r<s\}} \mathbb{E}_{B_{T_r}}[\mathbb{1}_{\{\sigma_{\{0\}}>0\}}]\big] = P_0(\sigma_{\{0\}} > 0)\, P_0(T_r < s) = 0.$$

Hence $P_0(J) = 0$. ∎

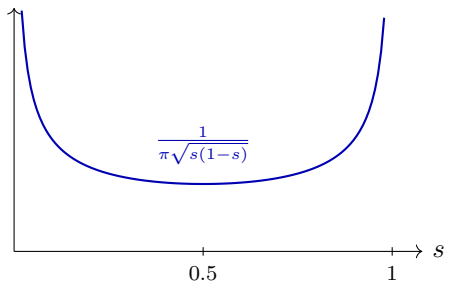


Figure. *Arcsine density of the last zero L before 1 (Example 5.22).*